STUDENT PRACTICAL EXAM STUDY GUIDE

Software Testing & Quality Assurance (STQA)

Practical Examination Objective

In the practical examination, students may be required to develop or modify a Selenium WebDriver test using Java.

Students should be able to:

1. Open the provided GitHub repository in GitHub Codespaces.
2. Check the Java, Maven, Chromium and ChromeDriver environment.
3. Navigate to the Maven project.
4. Understand the project structure.
5. Write/modify a Selenium WebDriver test.
6. Locate web elements.
7. Perform browser actions.
8. Validate expected results using assertions.
9. Execute the test using Maven.
10. Interpret the test result.

Important Technologies to Remember

Technology	**Purpose**
Java	Programming language
Selenium WebDriver	Browser automation
Chromium	Web browser
ChromeDriver	Connects Selenium with Chromium
Maven	Project build and dependency management
Junit	Testing framework
GitHub Codespaces	Online development environment

Practical No 1 : Creation of Test Suite Using Selenium IDE

Create a test suite containing four test cases for validating different web page formats (HTML, XML, JSON, and Dynamic Form) and execute the suite

HTML Page	https://example.com
XML Response	https://httpbingo.org/xml
JSON API	https://reqres.in/api/users?page=2
Dynamic Form	https://www.selenium.dev/selenium/web/web-form.html

Practical No 2 : Cross-Website Functional Testing Using Selenium IDE

Website 1

https://www.wikipedia.org

Website 2

https://the-internet.herokuapp.com

Practical No 3 : Manual Test Case Design and Execution

A website link will be provided during exam. Students will have to write 6 test cases, execute the test cases on the website given and record the actual result. Compare the actual and expected result and record the status. Also, create a defect report for the failed test cases.

*******NOTE – For all Selenium webdriver practicals, students have to memorise the 'STUDENT TASK' code only and update the java file accordingly during the practical exam. Thereafter, run the commands to execute the practical.**

However, the entire java file code is to be copied in your answer sheet *******

Practical No. 4: Login Automation and Validation using Selenium WebDriver (Java) with GitHub Codespaces

First Step — Open the Repository provided by the examiner

After opening the provided GitHub repository in Codespaces, check your current location.

Commands -

pwd

Expected - /workspaces/STQA-Exam-P4

Then display the files:

ls

You should see –

exam-pages

pom.xml

README.md

Src

Dockerfile

Check Java

Before starting the practical, check Java.

java -version

The examination repository uses **Java 21**.

Expected output will contain something similar to:

openjdk version "21..."

If Java is not installed

Check:

which java

If Java is missing, install it using:

sudo apt-get update

sudo apt-get install -y openjdk-21-jdk

Then verify:

java -version

Check Maven

Check Maven:

mvn -version

Expected output:

Apache Maven 3.8.7

The exact Maven version may vary.

The important point is that Maven should be available and should show Java 21.

If Maven is not installed

Run:

sudo apt-get update

sudo apt-get install -y maven

Then:

mvn -version

Check Chromium

Check:

which chromium

Expected:

/usr/bin/chromium

If it is not found, install:

sudo apt-get update

sudo apt-get install -y chromium

Then:

chromium --version

Check ChromeDriver

Check:

which chromedriver

Expected:

/usr/bin/chromedriver

If it is not found:

sudo apt-get update

sudo apt-get install -y chromium-driver

Then:

chromedriver --version

Check the java file. Update STUDENT TASKS (below code) in the java file

Task 1 -

driver.findElement(By.id("username")).sendKeys("student");

Task 2 -

driver.findElement(By.id("password")).sendKeys("1234");

Task 3 -

driver.findElement(By.id("loginBtn")).click();

To run the test -

mvn test

BUILD SUCCESS message along with Login result should be displayed.

Practical No 5 : WebElement Interaction Using Selenium WebDriver

(NO GUI)

In the **Terminal** at the bottom of the Codespaces window, run:

sudo apt-get update

Wait for the command to finish.

Then run:

sudo apt-get install -y maven chromium chromium-driver

Wait until the installation is complete.

Verify Maven

Run:

mvn -version

You should see output similar to:

Apache Maven 3.8.7

Java version: 21.0.8

Verify Chromium

Run:

chromium --version

You should see output similar to:

Chromium 150.0.7871.181

Verify ChromeDriver

Run:

chromedriver --version

You should see output similar to:

ChromeDriver 150.0.7871.181

COMPLETE THE STUDENT TASKS in the java file –

Task 1 -

```
WebElement nameInput = driver.findElement(By.id("name"));
nameInput.sendKeys("Conceicao");
```

Task 2 -

```
WebElement countryDropdown = driver.findElement(By.id("country"));
Select countrySelect = new Select(countryDropdown);
countrySelect.selectByVisibleText("India");
```

To run the test -

mvn test

Verify Successful Execution

A successful execution should contain:

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

The final result should be:

BUILD SUCCESS

Practical No. 6 – Data-Driven Testing using Selenium WebDriver and Apache POI

Step 1: Verify Java Installation

Open the Terminal and execute:

java -version

Expected output:

openjdk version "21"

Verify Maven Installation

Execute:

mvn -version

Expected output:

Apache Maven 3.9.x

If you receive:

mvn: command not found

install Maven:

sudo apt-get update

sudo apt-get install -y maven

Verify again:

mvn -version

Verify Git LFS

Execute:

git lfs version

If you receive:

git: 'lfs' is not a git command

install Git LFS:

sudo apt-get install -y git-lfs

git lfs install

Verify:

git lfs version

Verify Chromium Browser

Execute:

which chromium

If the command returns something similar to:

/usr/bin/chromium

Chromium is installed.

If there is **no output**, install Chromium:

sudo apt-get update

sudo apt-get install -y chromium chromium-driver

Verify Chromium Version

Execute:

chromium --version

Example output:

Chromium 151.x.x.x

Verify ChromeDriver

Execute:

which chromedriver

Expected output:

/usr/bin/chromedriver

If there is **no output**, install Chromium and ChromeDriver:

sudo apt-get update

sudo apt-get install -y chromium chromium-driver

Verify ChromeDriver Version

Execute:

chromedriver --version

Example output:

ChromeDriver 151.x.x.x

Open the Web Application Folder

In the terminal, execute:

cd webapp

Start the Local Web Server

Execute:

python3 -m http.server 8000

Expected output:

Serving HTTP on 0.0.0.0 port 8000...

Do not close this terminal.

Leave the server running.

COMPLETE THE STUDENT TASKS in the java file –

Task 1 -

driver.get("http://127.0.0.1:8000/index.html");

assertEquals("Student Management System", driver.getTitle());

Task 2 -

WebElement heading = driver.findElement(By.tagName("h1"));

assertEquals("Student Management System", heading.getText());

Open a New Terminal

From the VS Code menu:

Terminal → New Terminal

The first terminal should continue running the Python server.

Return to the Project Folder

In the new terminal, execute:

cd ..

Verify that you are back in the project folder:

pwd

Run the Selenium Test

Execute:

mvn test

Verify Successful Execution

The following messages should appear:

✓ Browser launched successfully.

✓ Student Management System opened successfully.

BUILD SUCCESS

Practical No 7 - Data Extraction and Score Analysis from Web Table

Verify the Project

Run:

```
pwd
ls
```

The project should contain files such as:

```
pom.xml
students.html
src/
```

Check Java and Maven

```
java -version
mvn -version
```

If Java or Maven Is Missing

```
sudo apt-get update
sudo apt-get install -y openjdk-21-jdk

java -version

For maven -

sudo apt-get update
sudo apt-get install -y maven

mvn -version
```

Check Chromium / Chrome

```
which chromium
which chromium-browser
which google-chrome
which chromedriver
```

If Chromium Is Missing

```
sudo apt-get update
sudo apt-get install -y chromium chromium-driver
```

Student task - Check whether student scored above 60 in at least one subject

```
        if (english > 60 ||
            mathematics > 60 ||
            science > 60) {

            studentsAbove60++;

            System.out.println("Status: Above 60");

        } else {

            System.out.println("Status: Below 60");

        }
    }
```

CRITICAL: Start the Web Server First

CRITICAL: DO NOT RUN mvn test BEFORE STARTING THE WEB SERVER.

In Terminal 1 run:

```
python3 -m http.server 8000
```

Open a Second Terminal

1. Leave Terminal 1 running.
2. Select Terminal → New Terminal.
3. Use Terminal 2 for Maven.

```
pwd
ls pom.xml
```

Run the Test

```
mvn test
```

BUILD SUCCESS